1. (A) Perform the Extraction, Transformation and Loading (ETL) Process in SQL Server

Aim

To perform **ETL (Extract, Transform, Load)** process to construct a **data warehouse database in SQL Server**.

Steps

1. Extraction

Extraction means **collecting data from different sources** like Excel, CSV files, or databases.

Example Source Table:

CustomerID	Name	City	Sales
101	Amit	Mumbai	5000
102	Riya	Pune	7000

In **SQL Server**, import data using:

Steps

1. Open **SQL Server Management Studio (SSMS)**
2. Right click **Database → Tasks → Import Data**
3. Select **Data Source (Excel/CSV)**
4. Select **Destination (SQL Server Database)**
5. Import the table.

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2. Transformation

Transformation means **cleaning and modifying the data**.

Examples:

- Removing null values
- Converting formats
- Aggregating data

Example SQL Transformation:

```
SELECT CustomerID, Name, City, Sales
FROM SalesData
WHERE Sales > 0;
```

Example Cleaning Data:

```
UPDATE SalesData
SET City = 'Mumbai'
WHERE City IS NULL;
```

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3. Loading

Loading means **storing transformed data into the Data Warehouse**.

Example:

```
INSERT INTO SalesWarehouse
SELECT * FROM SalesData;
```

Now the data is stored in the **data warehouse table**.

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1. (B) Apply What-If Analysis for Data Visualization

Aim

To perform **What-If Analysis and generate reports using Excel Pivot Tables and Charts**.

Steps

Step 1: Import Data

1. Open **Microsoft Excel**
2. Go to **Data → Get Data → From SQL Server**
3. Import the warehouse table.

Step 2: Create Pivot Table

1. Select the data
2. Click **Insert → Pivot Table**
3. Drag fields:

Rows → City
Values → Sales

Example Pivot Table:

City	Total Sales
Mumbai	15000
Pune	12000

Step 3: What-If Analysis

Use **Data → What-If Analysis → Scenario Manager**

Example Scenario:

Scenario	Sales Increase
Normal	10000
Growth	12000
High Growth	15000

Step 4: Create Pivot Chart

1. Click **Insert → Pivot Chart**
2. Select **Bar Chart**

This will visually represent the **sales analysis**.

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2. Perform Data Clustering using Python

Aim

To perform **data clustering using K-Means clustering algorithm in Python**.

Python Code Example

```
from sklearn.cluster import KMeans
import numpy as np

# Sample dataset
data = np.array([[5,3],
                 [10,15],
                 [15,12],
                 [24,10],
                 [30,45],
                 [85,70],
                 [71,80],
                 [60,78]])

# Apply K-Means
kmeans = KMeans(n_clusters=2)
kmeans.fit(data)

# Cluster centers
print(kmeans.cluster_centers_)

# Cluster labels
print(kmeans.labels_)
```

Output

The data points will be grouped into **2 clusters based on similarity**.

Example:

Cluster 1 → Small values
Cluster 2 → Large values